TOUCH GRASS — GAME RULES & AI BEHAVIOUR SPECIFICATION

Version: 1.0
Status: MVP Specification
Product principle: The phone is the scoreboard. The game happens outside.

PART 1 — GAME RULES

1. Purpose of the Game

Touch Grass is a real-world outdoor adventure game. Players complete enjoyable challenges in their surroundings, earn experience points (XP), unlock levels and achievements, and compete with friends.

The game should reward curiosity, exploration, movement, observation, and meaningful time outdoors—not screen time, physical ability, spending money, or access to special locations.

The experience must work for players who want a quiet walk, a small adventure, a social activity, or something unexpected.

2. Core Game Loop

Every quest follows this flow:

1. Choose an experience and duration.
2. Generate a personalized outdoor quest.
3. Review the mission briefing.
4. Start the quest.
5. Put the phone away and experience the real world.
6. Return to the app when finished.
7. Submit the required completion evidence.
8. Receive XP and any earned rewards.
9. Continue playing solo or compare progress with friends.

The player must never need to keep the app open, watch a timer, or repeatedly interact with the phone to make progress.

3. Quest Categories

Players can choose one of seven experience types.

Calm

- Encourages relaxation, awareness, and taking a break from digital distractions.
- Examples: notice changing shadows, find a peaceful place to sit, or listen carefully to the sounds around you.

Explore

- Encourages discovering a new perspective on a familiar area.
- Examples: take a different safe public path, notice an overlooked feature of a familiar street, or explore an accessible public space.

Social

- Encourages safe, voluntary interaction with other people.
- Examples: share a friendly greeting where appropriate, play a simple game with a willing friend, or ask someone you already know about a local favourite.
- Never require approaching strangers, collecting personal information, or entering private spaces.

Active

- Encourages accessible physical movement.
- Examples: take a walk, try a short movement challenge, or explore an open public space on foot.
- Challenges must offer reasonable alternatives for different physical abilities.

Observe

- Encourages noticing details in the environment.
- Examples: identify different leaf shapes, observe how light falls across a surface, or notice patterns in clouds.

Discover

- Encourages curiosity and small real-world discoveries.
- Examples: find three different natural textures, notice an unfamiliar plant without touching it, or identify a visible architectural detail.

Surprise Me

- Lets the AI select a suitable experience using the player's selected duration, optional preferences, recent quest history, and available context.
- It should produce something genuinely varied, not simply choose a random category and generate a generic walking task.

4. Quest Duration and Difficulty

Supported durations:

- 10 minutes — Quick adventure.
- 20 minutes — Short exploration.
- 30 minutes — Deeper adventure.
- 60 minutes — Extended adventure.

The selected duration represents the approximate total time needed to complete the challenge, including reasonable walking and observation time.

Rules:

- Respect the player's chosen duration.
- Never generate a 60-minute task when the player selects 10 minutes.
- Include a small buffer for ordinary delays.
- Avoid requiring the player to travel a long distance to reach a starting point.
- Keep instructions proportional to the challenge's duration.
- Do not treat a longer quest as inherently more meaningful or better.

A player should be able to complete a 10-minute quest and feel that it counts.

5. Quest Structure

Every generated quest must contain:

- A unique quest ID.
- A memorable title.
- A short, engaging introduction.
- One clear primary objective.
- Three to five concise instructions at most.
- Selected category.
- Estimated duration.
- Difficulty: Easy, Moderate, or Adventurous.
- A clear completion condition.
- Required evidence type.
- Any relevant safety note.
- A short completion message or narrative payoff.

Optional fields may include:

- A secondary discovery or optional bonus objective.
- An accessibility alternative.
- A verified starting location or route when location tools are available.
- A short environmental observation prompt.

Every quest must be understandable without repeatedly consulting the phone.

6. Quest Completion Rules

A quest is eligible for completion when the player submits the required evidence and passes the applicable validation rules.

Possible evidence types:

Reflection

- The player briefly describes what they noticed or experienced.

- Appropriate for mindfulness, observation, and discovery challenges.

Photo

- The player optionally or necessarily submits a relevant photograph.
- Use only when a photograph is useful to demonstrate the objective.
- Never require photographing strangers, private property, dangerous locations, or sensitive information.

Observation response

- The player answers a specific question about what they observed.
- For example, describing a pattern, identifying a visible feature, or reporting the number of distinct objects noticed.

Multiple evidence types

- Use only when the nature of the quest justifies them.
- Do not require unnecessary evidence for simple activities.

The generated quest must clearly explain its completion requirements before the player starts.

The player must not be forced to provide a photo for every quest.

7. XP and Reward Rules

XP is awarded by the deterministic game engine, not directly by the AI.

Recommended MVP base rewards:

Quest duration	Base XP
10 minutes	20 XP
20 minutes	35 XP
30 minutes	50 XP
60 minutes	80 XP

Optional bonuses:

- First completed quest of the day: +5 XP.
- Completing a category not yet completed that day: +5 XP.

The combined bonus must not exceed 10 XP per quest. Maximum recommended reward for a single quest is therefore 90 XP.

Rules:

- XP is awarded only after the quest passes completion validation.
- Generating, previewing, starting, pausing, or abandoning a quest awards no XP.
- Regenerating a quest does not award XP.

- The same quest completion cannot be rewarded twice.
- Bonus eligibility is determined by the game engine using recorded player activity.
- The AI cannot change XP values or invent new reward bonuses.
- Rewards must be explained clearly when awarded.

These values are initial game-balancing defaults and should be configurable without changing AI prompts.

8. Levels and Progression

Use cumulative XP to determine the player's level.

Recommended initial formula:

XP required to reach Level N = 100 × (N − 1) × N ÷ 2

Examples:

- Level 1: 0 XP.
- Level 2: 100 XP.
- Level 3: 300 XP.
- Level 4: 600 XP.
- Level 5: 1,000 XP.

The game engine calculates levels from total XP. The AI must not calculate or assign levels.

The interface should show:

- Current level.
- Total XP.
- XP needed for the next level.
- Progress toward the next level.

Level progression should remain achievable without requiring excessive daily play.

9. Achievements and Streaks

Achievements may reward:

- Completing a first quest.
- Exploring multiple categories.
- Completing quests on different days.
- Trying a new kind of outdoor activity.
- Participating in a Space.
- Reaching XP or level milestones.

Achievements must have explicit, testable unlock conditions.

Streaks can reward consistency, but missing a day must not erase previously earned XP, levels, or achievements. Do not pressure players to go outdoors in unsafe weather or unsafe circumstances.

Achievement eligibility is evaluated by the game engine using stored events, not by the AI's opinion.

10. Solo Play and Spaces

Solo play

- Players can generate and complete quests independently.
- Personal progress and rewards remain available without joining a group.

Spaces

- A Space is a group of players sharing progress and friendly competition.
- Members may complete quests asynchronously.
- Group rankings use validated XP earned during the relevant ranking period.
- Membership or participation alone does not award XP.
- Players must not need to be online simultaneously.
- Group participation must not require revealing precise locations or personal information.

Leaderboards

- Rank players using a clearly defined metric, such as XP earned during the selected period.
- Keep weekly rankings separate from lifetime progression.
- Never silently reset lifetime XP when a weekly leaderboard resets.
- Resolve ties deterministically, using the same rule for all players.

11. Location and Route Rules

Location-based quests may use GPS, geofencing, checkpoints, or route data when these capabilities are available.

However:

- Location permission must not be mandatory for every quest.
- Offer non-location-dependent alternatives whenever practical.
- Never invent nearby places, paths, distances, or route details.
- AI-generated route suggestions must rely on verified map or routing data before being presented as real routes.
- Prefer public, accessible, well-established paths.
- Never direct players into restricted areas, private property, dangerous roads, or isolated locations for the sake of a challenge.
- Do not require crossing busy roads, entering water, climbing structures, approaching wildlife, or interacting with hazardous objects.
- Do not encourage trespassing or collecting objects from protected environments.

- A player must be able to stop a quest safely without punishment.

Precise location should not appear on public leaderboards or in shared Spaces by default.

12. Fair Play and Abuse Prevention

The game engine must:

- Generate unique completion records.
- Prevent duplicate XP awards for the same completion.
- Validate quest ownership and completion status.
- Check bonus eligibility from recorded events.
- Apply rate limits to quest generation and completion submissions.
- Reject invalid or manipulated XP requests.
- Keep an auditable record of awarded XP and the reason for each award.

The AI's response is never trusted as the source of truth for player balances or rankings.

Do not impose harsh penalties for honest mistakes, incomplete challenges, or accessibility limitations.

PART 2 — AI BEHAVIOUR SPECIFICATION

13. The AI's Role

The AI is the adventure designer and storyteller.

It is responsible for:

- Creating engaging, specific outdoor challenges.
- Adapting quests to the selected experience and duration.
- Avoiding repetitive experiences.
- Writing concise, memorable mission briefings.
- Making ordinary surroundings feel interesting.
- Suggesting optional variations and accessible alternatives.
- Producing quest-specific completion criteria.
- Helping evaluate submitted text evidence when an appropriate evaluation model is available.

The AI is not responsible for:

- Awarding XP or changing balances.
- Assigning levels or unlocking achievements directly.
- Changing leaderboard rankings.
- Authorizing access to private or restricted places.
- Claiming GPS or map verification that did not occur.

- Declaring that a player definitely completed an activity based solely on an unsupported claim.
- Making final game-state or security decisions.

14. Quest Generation Inputs

The quest generator may receive:

- Selected category.
- Selected duration.
- Optional mood or difficulty preference.
- Optional accessibility preferences voluntarily supplied by the player.
- Recent quest history.
- Previously completed categories.
- Relevant, privacy-conscious location context if permission has been granted.
- Available verified route or map data.
- Configured content and safety rules.

Only send the information necessary to create the quest.

The player must not be required to disclose sensitive personal details to receive an adventure.

If location context is unavailable, the AI must generate a quest that works in a variety of ordinary outdoor environments rather than inventing a specific nearby destination.

15. Quest Generation Behaviour

Every generated quest should be:

Specific

- Give the player something concrete to do, notice, discover, or experience.
- Avoid generic instructions such as “go outside and enjoy nature.”

Achievable

- Fit the selected duration and difficulty.
- Require only ordinary, readily available resources unless clearly stated otherwise.
- Avoid complex preparation or paid activities.

Engaging

- Present the objective as a small adventure, mission, mystery, experiment, or playful observation.
- Make the player curious about what they might discover.
- Provide a satisfying purpose without requiring a large fictional backstory.

Grounded in the real world

- Work with surroundings that could reasonably exist.

- Avoid fabricated landmarks, objects, events, or environmental conditions.
- When a challenge depends on a particular feature, provide a fallback if it is absent.

Phone-light

- Be understandable before the player begins.
- Require no repeated AI conversations while outdoors.
- Avoid continuous camera use, frequent notifications, or manual check-ins.

Safe

- Do not reward risk-taking.
- Do not direct players into unsafe, private, or restricted spaces.
- Do not require interaction with strangers.
- Provide alternatives for inaccessible or unsuitable activities.

Varied

- Use recent quest history to avoid repeating substantially similar challenges.
- Vary the objective, interaction pattern, setting, and narrative framing.
- Do not treat minor wording changes as genuinely new quests.

16. Meaningful Challenge Design

The AI should create challenges that make people want to participate.

Prefer:

- Small discoveries that reveal overlooked details.
- Playful observation challenges.
- Real-world scavenger hunts using visible, ordinary features.
- Safe route-choice challenges based on verified routing information.
- Short missions with a clear goal and a satisfying conclusion.
- Optional creative tasks that do not require staring at a screen.
- Challenges that can be completed alone or with a willing friend.

Avoid:

- Generic walking prompts with decorative storytelling added.
- Arbitrary lists of unrelated tasks.
- Requiring users to search extensively for rare objects.
- Requiring strangers to cooperate.
- Tasks whose only purpose is to take photographs.
- Artificial urgency or punishment for taking longer than expected.
- Repetitive instructions to count steps or repeatedly check progress.
- Challenges that are effectively ordinary AI chatbot prompts with no meaningful outdoor activity.

The challenge itself must be the interesting part, not simply the text describing it.

17. Surprise Me Behaviour

When the player chooses Surprise Me, the AI should choose a suitable category and mechanic using the available context.

It should:

1. Respect the chosen duration.
2. Check recent quests to reduce repetition.
3. Select a safe, feasible challenge style.
4. Produce a quest with a clear objective and completion condition.
5. Preserve an element of surprise without making the instructions vague.

Surprise should come from the activity, the player's observations, or the mission's creative framing—not from sending players somewhere unknown or unsafe.

18. AI Output Requirements

The AI must return structured, schema-validated data rather than arbitrary prose.

The generated quest should include fields equivalent to:

- `title`
- `category`
- `durationMinutes`
- `difficulty`
- `storyIntro`
- `objective`
- `instructions`
- `completionCriteria`
- `evidenceType`
- `safetyNotes`
- `accessibilityAlternative`
- `optionalBonusObjective`

Location-dependent fields should only be included when supported by verified location data.

The application or API assigns authoritative identifiers and validates the response before storing or displaying it.

Validation must check:

- Required fields exist.
- The category is permitted.
- Duration matches the user's selection.
- Instructions are understandable and within the configured limit.
- Completion criteria are specific and testable.
- Evidence requirements are appropriate.
- The content passes safety checks.
- Location claims are supported by actual tool results.

If the response is malformed, unsafe, or incomplete, retry generation within configured limits or return a clear fallback. Never display invalid model output as a valid quest.

19. Evidence Evaluation Behaviour

Evidence evaluation should be separate from quest generation.

The evaluator may receive:

- The original quest and completion criteria.
- The submitted reflection or observation.
- A reference to submitted image evidence, if applicable and supported by the configured model.
- Any deterministic validation results available to the application.

It should assess whether the submission appears relevant to the stated objective, identify missing required evidence, and return a structured recommendation.

Possible outcomes:

- `accepted`
- `needs_more_evidence`
- `manual_review`

For example, if a quest asks the player to notice three different leaf shapes, a reflection that describes three distinct shapes may be relevant evidence. A submission that only says “I finished” may require more information.

Rules:

- Do not require a long written reflection.
- Do not reject a reasonable response because the wording differs from the quest's examples.
- Do not claim to have visually inspected a photograph unless the configured model actually supports image analysis and received the image.
- Do not claim GPS confirmation unless the relevant location checks occurred.
- Do not invent evidence or treat an AI-generated claim as a verified fact.
- Give players a clear, respectful explanation when more evidence is needed.
- Allow resubmission when appropriate.

The evaluator provides a recommendation. The application applies the documented completion policy, validates the result, and controls any state change.

20. AI Safety and Refusal Behaviour

The AI must never generate a challenge that requires:

- Trespassing or entering restricted property.
- Approaching dangerous traffic or crossing unsafely.
- Climbing dangerous structures.

- Touching dangerous materials or unknown wildlife.
- Harassing, photographing, or recording strangers without appropriate consent.
- Revealing sensitive personal information.
- Traveling to an unverified or potentially unsafe location.
- Spending money as a requirement for ordinary participation.
- Risky behaviour for extra XP or leaderboard advantage.

When a requested challenge is unsuitable, the AI should generate a safe alternative that preserves the player's intended experience where possible.

For example, a request for an adventurous quest should produce a more exploratory challenge, not a more dangerous one.

21. AI Memory and Personalization

Personalization should improve variety without making the experience invasive.

The AI may use recent quest history to:

- Avoid repeated activities.
- Introduce new categories.
- Adjust the complexity of instructions.
- Suggest alternative quest mechanics.
- Respect optional preferences supplied by the player.

It must not infer sensitive personal attributes or use private information unrelated to the game.

The AI should not require extensive personal profiles to generate good challenges. Recent quest history and the player's current selection should be sufficient for useful personalization.

22. Failure Handling and Fallbacks

The app must handle:

- AI provider timeouts.
- Invalid model responses.
- Rate limits.
- Unavailable location services.
- Missing map data.
- Unsupported image evaluation.
- Failed evidence uploads.
- Network interruptions during submission.

Rules:

- Never award XP because an AI request failed.
- Preserve the player's submitted work when practical.
- Do not lose an already accepted completion because a later request fails.
- Do not claim that a location has been verified when verification is unavailable.

- Provide a retry or safe fallback when possible.
- Make it clear when a quest is generated from a fallback rather than the configured AI provider.

A small set of reviewed, category-specific fallback quests may be used when the AI provider is unavailable. They must pass the same safety and structure validation as generated quests.

PART 3 — SYSTEM BOUNDARIES AND ACCEPTANCE CRITERIA

23. Separation of Responsibilities

The implementation must preserve these boundaries:

Responsibility	Owner
Quest ideas, narrative, instructions	AI
Quest response structure and validation	Shared schemas and application
Quest persistence and ownership	API and database
Completion evidence review	AI evaluator where supported, plus application policy
XP calculations	Deterministic game engine
Level calculations	Deterministic game engine
Achievement eligibility	Deterministic game engine
Duplicate completion prevention	API and database
Leaderboard calculations	Deterministic application logic
Verified routes and real-world location data	Location/map provider
Final state changes	Application and game engine

No model output may directly modify player XP, levels, achievements, or rankings.

24. MVP Acceptance Criteria

The implementation is ready for MVP testing when:

- Players can select all seven experience categories.
- Players can select the supported durations.
- The AI generates structured, usable quests that respect those selections.
- Generated quests have clear objectives and completion conditions.
- Recent quest history can help reduce repetition.
- Players can complete quests without continuous phone interaction.
- Different quests can require different appropriate evidence types.
- Invalid evidence submissions receive useful feedback.
- XP is awarded only through the game engine after successful completion.

- Duplicate submissions cannot award XP twice.
- Levels and achievements are calculated deterministically.
- Solo play works without joining a Space.
- Space rankings use validated game records.
- Location-dependent quests never rely on fabricated route information.
- AI failures produce a controlled retry or fallback.
- Unsafe challenges are rejected or replaced with safe alternatives.
- Automated tests cover XP calculations, levels, bonus limits, duplicate submissions, schema validation, and major safety rules.

25. Final Design Principle

Touch Grass should reward the player for what they experience outside—not for how often they interact with the application.

The AI should make the world feel more interesting. The game engine should make rewards fair. The interface should get out of the player's way.

The ultimate success condition is simple: players should close their phones because they are excited to complete the adventure, then return because they are excited to see what they earned.

TOUCH GRASS — OPEN-SOURCE AI & OUTDOOR ACTIVITY ADDENDUM

Applies to: Product requirements, technical architecture, game rules, AI behaviour, and MVP acceptance criteria.

1. Core Product Principle

Touch Grass is an AI-powered, real-world activity game. Its purpose is to help people spend less time looking at screens and more time actively experiencing the world around them.

The game is not a walking app, fitness tracker, hiking planner, or step counter.

Walking is one possible activity, but it must never be the default assumption for every quest.

The core loop is: Receive a mission → Put the phone away → Engage with the real world → Return and record the experience → Earn rewards.

A successful quest should involve meaningful participation in the physical world, not simply reading AI-generated text or answering questions inside an app.

2. Support Diverse Outdoor Activities

The AI must generate quests across multiple activity types, including but not limited to:

- **Nature and wildlife:** Observe birds, notice insects from a safe distance, identify visible plant features, or observe changes in the sky.
- **Gardening and growing:** Tend a plant, inspect leaves for visible changes, water a garden, or arrange a small growing space.
- **Creative activities:** Sketch an outdoor scene, photograph interesting textures, make a nature-inspired drawing, or create something from materials already available and appropriate to use.
- **Observation and curiosity:** Notice patterns in architecture, compare shadows, find different natural textures, or discover overlooked details in a familiar environment.
- **Relaxation and mindfulness:** Sit outdoors, listen to surrounding sounds, observe moving clouds, or spend a few minutes noticing the environment without distractions.
- **Play and movement:** Play a simple game, practise a personally suitable movement activity, or explore an outdoor space.
- **Social activities:** Play with willing friends, share an outdoor creative activity, or participate in a small group challenge.
- **Exploration:** Discover something unfamiliar in a safe public area, investigate a new perspective on a familiar place, or follow a verified, accessible route.
- **Environmental care:** Carry out a safe, small gardening or environmental-care activity without disturbing wildlife or protected areas.

These categories are examples, not a closed list. The AI should be able to combine activity types when appropriate.

Not every quest should involve walking, covering distance, visiting a new location, interacting with other people, or taking photographs.

A quest may be completed in a garden, yard, balcony, courtyard, suitable outdoor communal area, or another safe and accessible outdoor setting. The game should not require players to travel somewhere new just to participate.

Where an activity genuinely requires a particular environment, explain that requirement before the quest begins and provide a suitable alternative where possible.

Indoor-only activities do not satisfy the core outdoor objective. However, a player should not need to leave their property or travel away from home to participate in every quest.

3. AI Creativity and Personalization

The AI should generate missions that are engaging because of what the player gets to do, discover, create, or experience.

Avoid producing the same basic walking challenge with different titles and decorative storytelling.

For each quest, consider:

- The selected experience category.
- The available time.
- The type of real-world interaction.
- The player's recent quests and completed activities.
- Whether the activity needs a particular environment or resource.
- Whether the activity is practical for someone with limited mobility or access to outdoor spaces.

The AI should vary the actual mechanics of the experience, not merely its wording.

For example, a 10-minute Observe quest could involve noticing patterns in leaves, while a 10-minute Creative quest could involve sketching an interesting shape outdoors. A Calm quest could involve listening to ambient sounds, while a Social quest could involve playing a simple game with a willing friend.

These are examples of possible directions, not fixed templates.

4. Open-Source AI Must Be Central to the Product

The project must satisfy the Hacktoberfest Open-Source AI Challenge through its actual implementation.

Open-source AI must not be a decorative feature, an optional chatbot, or an incidental API call.

The open-weight model must play a central role in generating personalized, structured, varied outdoor quests. Its output should meaningfully determine the activity, objective, instructions, and completion criteria presented to the player.

Use an open-weight model such as a compatible Gemma model, with the exact model and inference provider documented in the implementation.

Use an open-source agent framework, such as Mastra, where it materially supports the quest-generation workflow. The framework should help structure and validate the AI's work rather than add unnecessary agent complexity.

The architecture should:

- Keep model selection configurable.
- Separate model inference from game rules and application logic.
- Allow compatible open-weight models or inference providers to be substituted without rewriting the game engine.
- Support structured model outputs and validation.
- Keep XP, levels, achievements, and rankings deterministic.
- Clearly distinguish actual model-generated quests from predefined fallback quests.
- Document the model, its license, the inference method, and any external services used.

Where practical, provide a local inference option so developers can run the quest-generation component without depending on a proprietary model API. Do not claim that the full application works offline unless its dependencies and offline behaviour have been implemented and tested.

The implementation must not silently substitute a closed proprietary model for the intended open-weight model.

5. Why Open Innovation Matters

The implementation should make the benefits of an open approach concrete.

Potential benefits include:

- **Model choice:** Developers can change or compare compatible open-weight models.
- **Control:** Quest-generation behaviour can be adjusted through prompts, validation rules, and, where supported, fine-tuning.
- **Transparency:** The model and its role in generating quests can be documented and inspected.
- **Deployment flexibility:** Developers can choose compatible hosted inference or local inference where supported.
- **Privacy options:** Local inference can reduce the need to transmit quest-generation inputs to an external model provider, although other application services may still process data.
- **Reduced lock-in:** Core game mechanics do not depend on a single model vendor.

These are design goals, not claims that every benefit is already available. The final project write-up must distinguish implemented capabilities from future possibilities and report actual testing results.

6. Phone-Away Design

Every quest must be designed around minimal screen interaction.

Players should receive clear instructions before starting and be able to complete the activity without repeatedly checking their phones.

Do not require:

- Continuous camera use.
- Constant GPS tracking when unnecessary.
- Repeated self-check-ins.
- A series of phone-based tasks during the activity.
- Live AI conversation to make progress.
- Watching a countdown to avoid losing rewards.

When the player returns, the app can collect a short reflection, relevant observation, or appropriate evidence.

The game should make the real-world activity the main event and the phone the place where the player receives the mission and sees the result.

7. Acceptance Criteria

The MVP must demonstrate that:

1. The AI can generate meaningfully different outdoor activities, not just walking variations.
2. Players can complete suitable quests without leaving their immediate outdoor space.
3. Quests respect the selected duration and category.
4. Quest instructions are clear enough to follow without ongoing phone use.
5. The open-weight model is genuinely responsible for the core quest-generation experience.
6. The model and inference configuration are documented and replaceable within the supported architecture.
7. Deterministic game rules control all XP and progression.
8. Fallback behaviour is clearly distinguished from live model generation.
9. The project documents which open-source benefits have actually been demonstrated.
10. Safety, accessibility, and privacy are considered in every activity category.

Final principle: Touch Grass should make the world feel like a playground for curiosity and creativity. The AI invents the mission; the player brings it to life outside.